

INFERENCEAL STATISTICAL TESTS

Hypothesis Testing (Comparison Tests)

- **t-Test (Independent & Paired)** – Comparing means of two groups (A/B testing, before-after analysis).
- **ANOVA (One-Way & Two-Way)** – Comparing means of multiple groups (customer segmentation, experimental design).
- **Z-Test** – Comparing sample mean to population mean (large sample size).

1. t-Test (Independent & Paired) – Detailed Explanation & Real-World Example

What is a t-Test?

A **t-test** is a statistical test used to compare the means of two groups to determine if the difference between them is statistically significant. It helps answer questions like:

- Is there a significant difference in customer spending before and after a marketing campaign?
- Do two different website designs lead to different conversion rates?

The t-test assumes that the data follows a **normal distribution** and that variances are equal in the two groups (for independent samples). If these assumptions are violated, non-parametric tests (like the Mann-Whitney U test) are better alternatives.

Types of t-Tests

1. Independent Samples t-Test (Two-Sample t-Test)

- ✓ **Use case:** Compare the means of two independent groups.
- ✓ **Assumption:** The two samples are unrelated (e.g., different customer segments, different product groups).

✓ Formula:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- $\bar{X}_1, \bar{X}_2$ are the sample means
- s_1^2, s_2^2 are the variances
- n_1, n_2 are the sample sizes

2. Paired Samples t-Test (Dependent t-Test)

✓ **Use case:** Compare means before and after an intervention (e.g., impact of a training program).

✓ **Assumption:** The two samples are related (e.g., same individuals before and after an event).

✓ Formula:

$$t = \frac{\bar{D}}{s_D / \sqrt{n}}$$

where:

- $\bar{D}$ = mean of the differences between paired observations
- s_D = standard deviation of differences
- n = sample size

- ✓ Use an **Independent t-Test** for comparing two different groups (e.g., A/B testing).
- ✓ Use a **Paired t-Test** for comparing before-and-after situations (e.g., impact analysis).
- ✓ In real-world business, these tests help in decision-making, A/B testing, and assessing marketing impact.

(A). Independent t-Test (A/B Testing for Website Conversion)

Scenario: A company wants to test if a new website design (Version B) improves conversion rates compared to the old design (Version A).

Step 1: Define Hypotheses

- Null Hypothesis (H0): There is no significant difference in conversion rates between Version A and Version B.
- Alternative Hypothesis (H1): Version B has a significantly higher conversion rate.

$$1 - 0.05 = 0.95$$

$$\alpha = 0.05$$

Step 2: Given Data

Version	Sample Size (n)	Mean Conversion Rate ($\bar{x}$)	Standard Deviation (s)
A (Old)	500	3.2% (0.032)	1.1% (0.011)
B (New)	500	4.1% (0.041)	1.3% (0.013)

Step 3: Calculate the t-Statistic

The formula for an Independent t-Test is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where:

- $\bar{X}_1, \bar{X}_2$ = sample means
- s_1^2, s_2^2 = sample variances
- n_1, n_2 = sample sizes

Step 3.1: Compute Variance for Each Group

$$s_1^2 = (0.011)^2 = 0.000121$$

$$s_2^2 = (0.013)^2 = 0.000169$$

Step 3.2: Compute Standard Error (SE)

$$\begin{aligned}
 SE &= \sqrt{\frac{0.000121}{500} + \frac{0.000169}{500}} \\
 &= \sqrt{0.000000242 + 0.000000338} \\
 &= \sqrt{0.00000058} \\
 &= \underline{0.00076}
 \end{aligned}$$

Step 3.3: Compute t-Statistic

$$t = \frac{0.032 - 0.041}{0.00076}$$

$$= \frac{-0.009}{0.00076}$$

$$= -11.84$$

Step 4: Find the Critical t-Value

- Degrees of freedom (df): $df = n_1 + n_2 - 2 = 500 + 500 - 2 = 998$
- From a t-table, for $df \approx 1000$ at $\alpha = 0.05$ (two-tailed test), the critical t-value is ± 1.96 .

Step 5: Compare t-Value with Critical t-Value

- Computed $t = -11.84$
- Critical $t = \pm 1.96$
- Since $|t| > 1.96$, we reject H_0 .

Step 6: Conclusion

Since $p < 0.05$, there is a statistically significant difference in conversion rates. The company should implement Version B because it has a significantly higher conversion rate.

(B). Paired t-Test (Sales Before vs. After Marketing Campaign)

Scenario:

A company wants to see if a **marketing campaign** significantly increases customer sales.

Step 1: Define Hypotheses

- Null Hypothesis (H0):** No significant difference in customer sales before and after the campaign.
- Alternative Hypothesis (H1):** The campaign increased customer sales.

Step 2: Given Data

Customer	Sales Before (\$X_1\$)	Sales After (\$X_2\$)	Difference (\$D = X_2 - X_1\$)
1	200	250	50
2	150	180	30
3	300	320	20
4	180	200	20

Customer	Sales Before (\$X_1\$)	Sales After (\$X_2\$)	Difference (\$D = X_2 - X_1\$)
5	250	290	40

Step 3: Compute Mean and Standard Deviation of Differences

Step 3.1: Compute Mean Difference ($\bar{D}$)

$$\bar{D} = \frac{50 + 30 + 20 + 20 + 40}{5}$$

$$\bar{D} = \frac{160}{5} = \underline{32}$$

Step 3.2: Compute Standard Deviation of Differences (s_D)

First, find squared differences from the mean:

$$\sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

Customer	Difference (\$D\$)	$(D - \bar{D})^2$
1	50	$(50 - 32)^2 = 324$ -
2	30	$(30 - 32)^2 = 4$ -
3	20	$(20 - 32)^2 = 144$ -
4	20	$(20 - 32)^2 = 144$ -
5	40	$(40 - 32)^2 = 64$ -

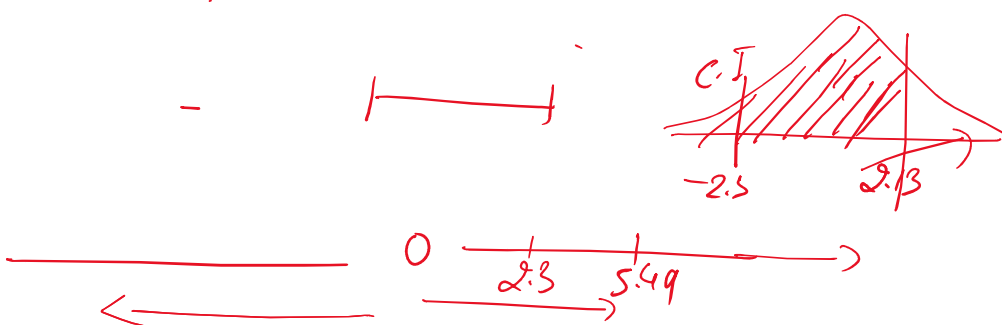
$$s_D^2 = \frac{324 + 4 + 144 + 144 + 64}{5 - 1}$$

$$s_D^2 = \frac{680}{4} = 170$$

$$s_D = \sqrt{170} \approx 13.04 \checkmark$$

t-test $\rightarrow$ 11 | 5.49

t-table $\rightarrow$ ± 1.96 2.13



Step 4: Compute t-Statistic

The formula for a Paired t-Test is:

$$t = \frac{\bar{D}}{s_D / \sqrt{n}}$$

Substituting values:

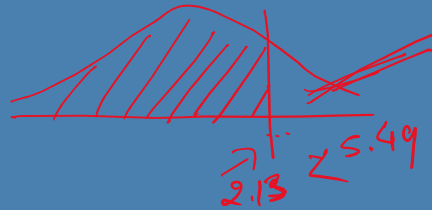
$$\begin{aligned} t &= \frac{32}{13.04 / \sqrt{5}} \\ &= \frac{32}{13.04 / 2.236} \\ &= \frac{32}{5.83} \\ &= 5.49 \end{aligned}$$

Step 5: Find the Critical t-Value

- Degrees of freedom (df) = $n - 1 = 5 - 1 = 4$
- From a t-table, for df = 4 at $\alpha = 0.05$ (one-tailed test), the critical t-value is 2.13

Step 6: Compare t-Value with Critical t-Value

- Computed $t = 5.49$
- Critical $t = 2.13$
- Since $5.49 > 2.13$, we reject H_0 .



Step 7: Conclusion

Since $p < 0.05$, the marketing campaign significantly increased customer sales, and the company should continue running similar campaigns.

2. Z-Test: Detailed Explanation & Real-World Industrial Example

What is a Z-Test?

A Z-Test is a statistical test used to compare a sample mean to a population mean when the sample size is large ($n \geq 30$). It determines whether the observed difference is statistically significant or due to random chance.

When to Use a Z-Test?

- ✓ Population standard deviation (σ) is known
- ✓ Sample size is large ($n \geq 30$)
- ✓ Data is normally distributed (or follows the Central Limit Theorem)

If the sample size is small ($n < 30$) and population standard deviation is unknown, use a t-Test instead.

Types of Z-Tests

1. ✓ **One-Sample Z-Test:** Compares the sample mean to a known population mean.
2. ✓ **Two-Sample Z-Test:** Compares the means of two independent samples.

(A). One-Sample Z-Test: Manual Calculation & Example

Industry Example: Average Salary of Data Analysts

A company claims that the average salary of a data analyst in New York is \$85,000 per year. A researcher collects a random sample of 50 data analysts and finds the average salary is \$82,000, with a known population standard deviation of \$12,000.

We want to test if the average salary in the sample is significantly different from the claimed \$85,000.

Step-by-Step Calculation of One-Sample Z-Test

Step 1: Define Hypotheses

- Null Hypothesis (H_0): The population mean is \$85,000 ($\mu = 85,000$)
- Alternative Hypothesis (H_1): The population mean is not \$85,000 ($\mu \neq 85,000$)

This is a two-tailed test since we are checking for any difference (higher or lower).

Step 2: Given Data

- Population Mean (μ) = 85,000
- Sample Mean ($\bar{X}$) = 82,000
- Population Standard Deviation (σ) = 12,000
- Sample Size (n) = 50

Step 3: Compute the Z-Statistic

The Z-Test formula is:

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Step 3.1: Compute Standard Error (SE)

$$\alpha = \frac{0.05}{2} = 0.025$$

$$1 - 0.025 = 0.975$$

$$1.9 + 0.06$$

$$\pm 1.96$$

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$= \frac{12,000}{\sqrt{50}}$$

$$= \frac{12,000}{7.07}$$

$$= 1,697.06$$

Step 3.2: Compute Z-Score

$$Z = \frac{82,000 - 85,000}{1,697.06}$$

$$Z = \frac{-3,000}{1,697.06}$$

$$Z = -1.77$$

$$\leftarrow -1.96 \quad \text{-----} \quad 1.96$$

Step 4: Find the Critical Z-Value

- Significance Level (α) = 0.05 (for a 95% confidence level)
- For a two-tailed test, the critical Z-values are ± 1.96 .

Step 5: Compare Z-Statistic with Critical Value

- Computed $Z = -1.77$
- Critical $Z = \pm 1.96$

Since $|Z| < 1.96$, we fail to reject H_0 .

Step 6: Conclusion

There is not enough evidence to say that the average salary of data analysts is different from \$85,000. The difference is likely due to random variation.

(B). Two-Sample Z-Test: Manual Calculation & Example

$1 - 0.05$
 $\Rightarrow 0.9500$

Industry Example: Customer Satisfaction in Two Cities

A retail company wants to compare customer satisfaction scores between New York and Los Angeles. They collect random samples from both cities.

- New York: $\bar{X}_1 = 7.8, n_1 = 100, \sigma_1 = 1.2$
- Los Angeles: $\bar{X}_2 = 7.3, n_2 = 120, \sigma_2 = 1.4$
- Population standard deviations are known

We want to test if customer satisfaction is significantly different between the two cities.

Step-by-Step Calculation of Two-Sample Z-Test

Step 1: Define Hypotheses

- Null Hypothesis (H_0): There is no difference in customer satisfaction ($\mu_1 = \mu_2$)
- Alternative Hypothesis (H_1): Customer satisfaction is different ($\mu_1 \neq \mu_2$)

This is a two-tailed test.

Step 2: Given Data

City	Sample Mean ($\bar{X}$)	Sample Size (n)	Standard Deviation (σ)
New York	7.8	100	1.2
Los Angeles	7.3	120	1.4

Step 3: Compute the Z-Statistic

The Two-Sample Z-Test formula is:

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step 3.1: Compute Standard Error (SE)

$$\begin{aligned} SE &= \sqrt{\frac{(1.2)^2}{100} + \frac{(1.4)^2}{120}} \\ &= \sqrt{\frac{1.44}{100} + \frac{1.96}{120}} \\ &= \sqrt{0.0144 + 0.0163} \\ &= \sqrt{0.0307} \\ &= 0.175 \end{aligned}$$

Step 3.2: Compute Z-Score

$$\alpha = \frac{0.05}{2} = 0.025$$

$$1 - 0.025 = 0.975$$

$$\begin{aligned} Z &= \frac{7.8 - 7.3}{0.175} \\ &= \frac{0.5}{0.175} \\ &= \underline{2.86} \end{aligned}$$

Step 4: Find the Critical Z-Value

- Significance Level (α) = 0.05
- Two-tailed test, critical Z-values are ± 1.96

Step 5: Compare Z-Statistic with Critical Value

- Computed $Z = \underline{2.86}$
- Critical $Z = \pm 1.96$
- Since $Z = 2.86 > 1.96$, we reject H_0 .



Step 6: Conclusion

There is a **significant difference** in customer satisfaction scores between New York and Los Angeles. The company should investigate further.

Final Summary

Test	Use Case	Decision
One-Sample Z-Test	Compare sample mean to population mean	Fail to reject H_0 (No significant difference)
Two-Sample Z-Test	Compare two independent sample means	Reject H_0 (Significant difference found)